GUESS for Text, Knowledge and Citation Networks

Character and word networks from text, citation analysis and main paths, co-authorship, keyword maps, document similarity and knowledge graphs

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Why networks?

Texts and the literature about them are full of implicit networks. Characters share scenes, words share sentences, papers cite papers, authors write together, and documents share vocabulary. Making these networks explicit answers questions such as:

- Who is at the centre of a story, and who connects its worlds?
- What are the themes of a corpus, and how do they relate?
- Where did a research field come from, and how is it organised?

This book assumes the Kitchen for loading, querying and drawing. Its data are in the cookbooks' `data` folder:

file	what it is
`scandal-in-bohemia.txt`	Arthur Conan Doyle's "A Scandal in Bohemia" (1891), the first Sherlock Holmes short story. Public domain, from Project Gutenberg.
`vis-papers.gdf`	1,044 highly cited papers whose titles mention graph drawing, graph layout or network/graph visualisation (1980–2024), with the 4,073 citations among them (citing → cited). Node fields: `title`, `year`, `cites` (citations from anywhere), `firstauthor`, `venue`. From OpenAlex, CC0.
`vis-papers-nodes.csv`	the same papers as a node table
`vis-paper-authors.csv`	paper, author: one row per authorship
`vis-paper-keywords.csv`	paper, keyword, score: OpenAlex's keywords for each paper

Networks from text

Map the characters of a story Basic

You have a story. **You want** its cast as a network: who appears with whom.

Ingredients: `scandal-in-bohemia.txt` · `makeFromText`

```
cast = {"Holmes": ["Holmes", "Sherlock"], "Watson": ["Watson", "Doctor"],
        "Irene Adler": ["Irene Adler", "Irene", "Miss Adler", "the woman", "Mrs.
         ↪ Norton"],
        "the King": ["King", "Majesty", "Count Von Kramm", "Von Kramm"],
        "Godfrey Norton": ["Godfrey Norton", "Godfrey", "Mr. Norton"],
        "John": ["John"], "Mrs. Turner": ["Mrs. Turner"]}
makeFromText("scandal-in-bohemia.txt", cast)
sizeBy(count, 10, 50)
sizeBy(weight, 1, 10)
labelBy(label)
circleLayout()
```

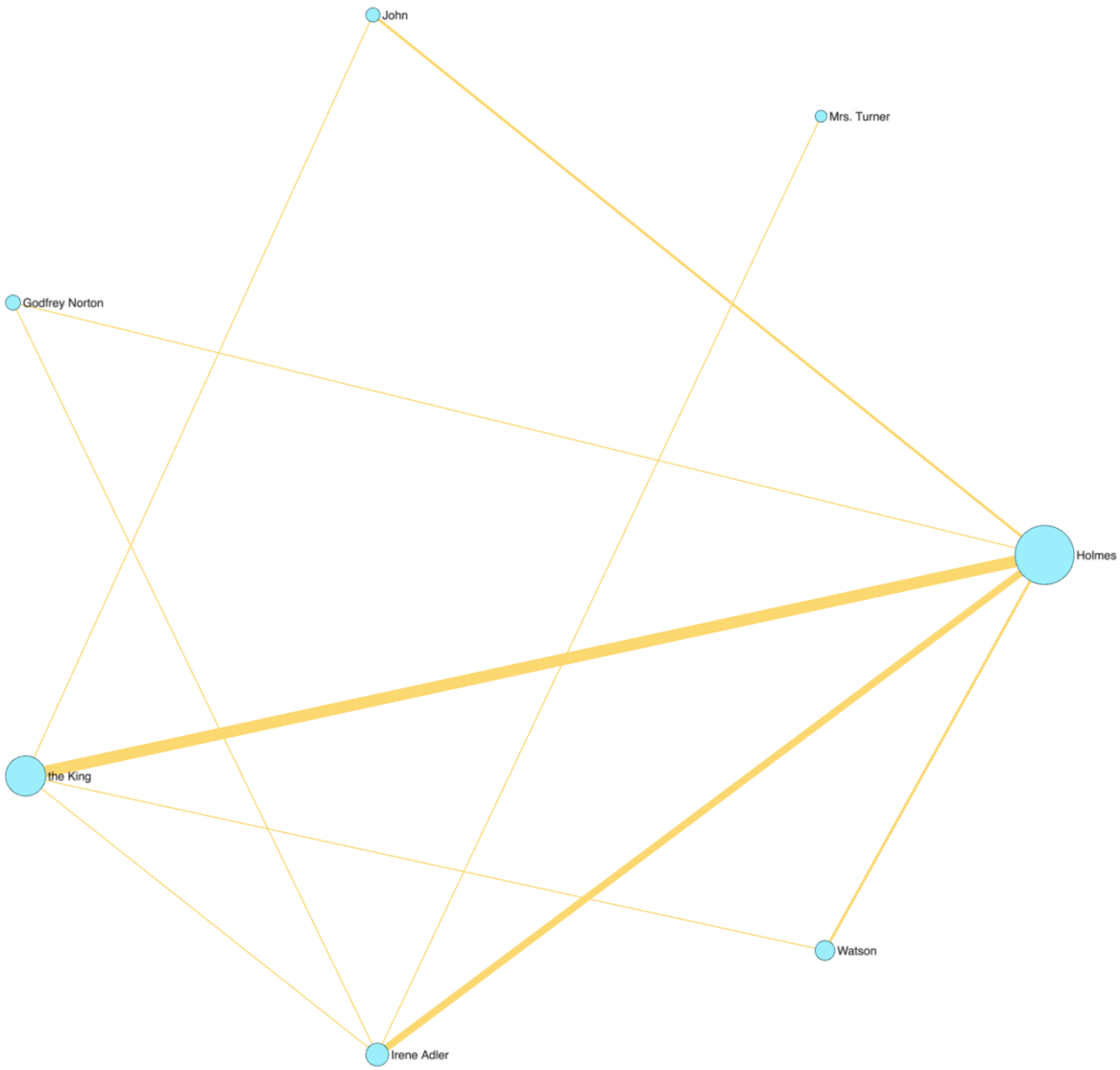


Figure 1: Who shares paragraphs with whom in "A Scandal in Bohemia"

How it works. `makeFromText(file, terms)` splits the text into paragraphs and links two characters when they're named in the same one. `weight` counts those paragraphs. Each character is a key in the dictionary, and its list holds every way the text refers to them, so "Irene Adler", "Miss Adler" and "the woman" all count for one node. Nodes get `count` (mentions), `units` (paragraphs) and `first` (the first paragraph they appear in). The strongest tie is Holmes and the King (8 paragraphs), then Holmes and Irene Adler (5). The story is about the contest between them, conducted through a client.

Variations. `by="sentence"` uses smaller units, giving fewer, more intimate links. `window=15` links

names within 15 words of each other. `scatterLayout(first, count)` puts each character at the point where they enter the story, which gives a rough cast timeline.

Pitfalls. Mentions aren't presence. Watson is named only 11 times because he narrates, as "I". A first-person narrator is everywhere and nowhere in a co-occurrence network. Pronouns ("he", "she") aren't counted at all. Treat the network as "who is named together".

Build a word network Intermediate

You have text with no list of names. **You want** its main words and how they cluster.

```
makeFromText("scandal-in-bohemia.txt", maxTerms=50, minCount=4,
             stopwords=["cried", "come", "came", "asked", "looked", "know",
 ↪  "see"])
remove(weight < 2)
louvainClusters()
colorize(modularity_class)
sizeBy(count, 6, 30)
labelBy(label)
forceAtlas2Layout()
```

How it works. Without `terms`, `makeFromText` takes the `maxTerms` most frequent words that occur at least `minCount` times, ignoring a built-in list of common English words (*the*, *of*, *said*…) and any `stopwords` you add. `remove(weight < 2)` drops pairs that co-occur only once, the usual first step against a hairball. Communities then group words that keep company.

Pitfalls. In a single short story, frequent words are mostly the story's furniture (*room*, *door*, *street*), and communities are weak (modularity around 0.2). Word networks earn their keep on larger corpora (many abstracts, many speeches), with a domain stopword list and a stricter weight threshold.

Citation networks

Explore a citation network Basic

Ingredients: `vis-papers.gdf` · `indegree` · `pagerank` · `generation`

```
makeFromGDF("vis-papers.gdf")
removeDisconnected()
for n in sortBy(indegree)[-5:]: print(n.indegree, n.cites, n.label, n.title[:50])
for n in sortBy(pagerank)[-5:]: print(round(n.pagerank, 3), n.label,
 ↪  n.title[:50])
```

How it works. Edges run from the citing paper to the cited one, so `indegree` counts citations *within this set*. `cites` is the paper's citation count from anywhere, which is much larger. Fruchterman and Reingold's 1991 force-directed paper is the most cited here. `pagerank` rewards papers cited by well-cited papers, so it favours foundations: the annotated bibliography of Di Battista et al. (1994) and Eades's 1984 spring heuristic rank highest. `generation` is each paper's topological level, counting back through chains of citation. `removeDisconnected()` drops the 265 papers with no citation links to the rest of the set.

Find the main path of a field Intermediate

You want the backbone of a field's history: the chain of papers through which most of its citation paths run.

```
makeFromGDF("vis-papers.gdf")
removeDisconnected()
mp = mainPath()
showOnly(mp)
g.edges.visible = false
mp.visible = true
scatterLayout(year, generation)
mp.color = red
labelBy(label)
g.nodes.labelpos = "outside"
optimizeLabels()
```

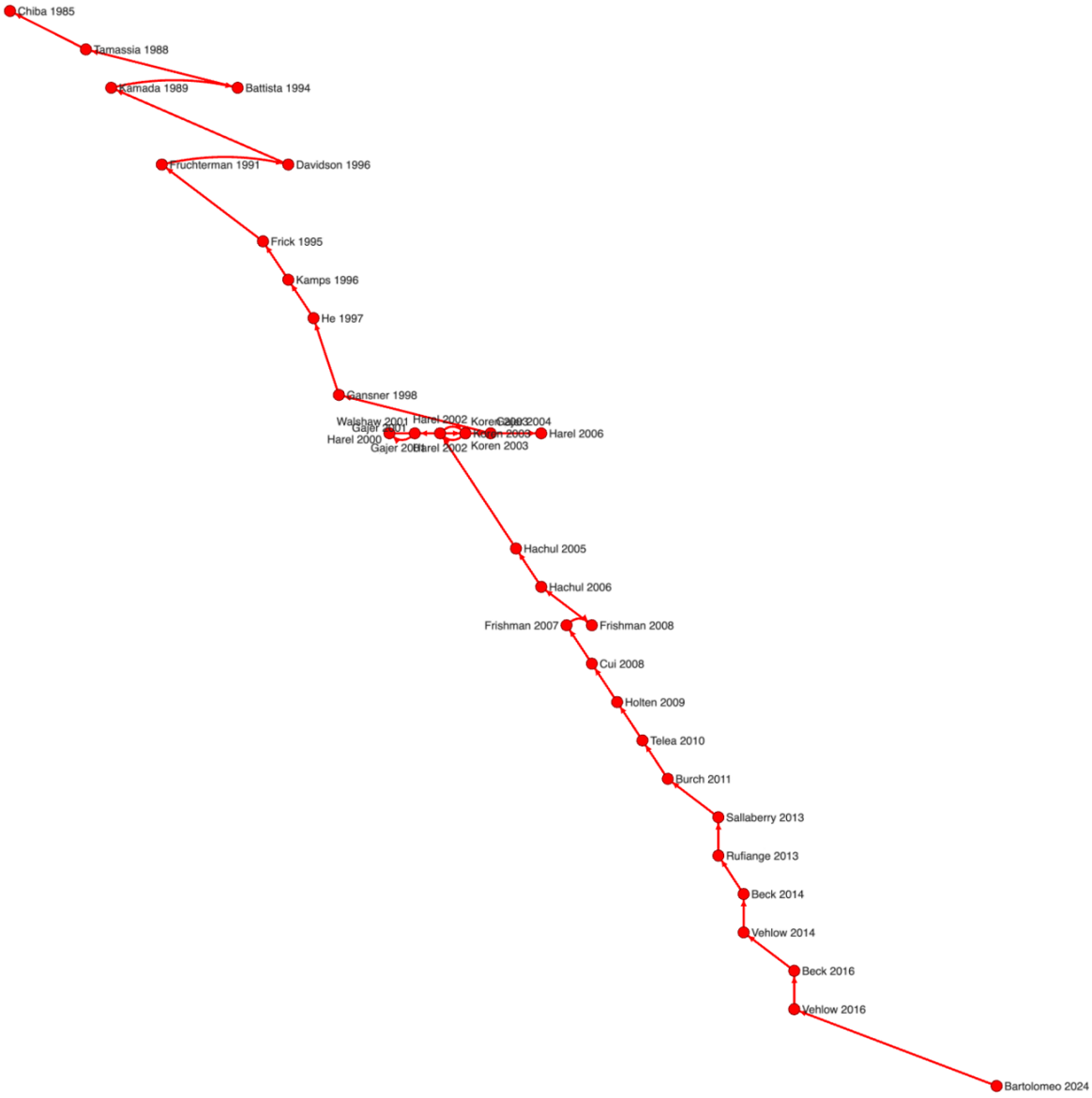


Figure 2: The main path of graph-drawing research

How it works. `mainPath()` performs main path analysis (Hummon and Doreian). For every citation it counts how many chains from the newest papers back to the oldest pass through it (its *search path count*, stored as `spc`, a share of all such chains). Then it follows the heaviest links. `showOnly(mp)` keeps the path's papers, and `mp.visible = true` (after hiding every edge) keeps only its citations. `scatterLayout(year, generation)` lays it out as a timeline: year across, citation depth up. Here the path runs from Chiba et al. (1985) and Tamassia et al. (1988) through Kamada–Kawai (1989) and Fruchterman–Reingold (1991) to the multilevel methods of 2000–2006 (Walshaw, Harel and Koren, Hachul and Jünger). It then turns to edge bundling (2008–2010, including Holten's force-directed bundling), then to the visualisation of dynamic graphs (2011–2016), and ends at a 2024 review of how layout algorithms are evaluated. That's the field's arc from drawing small graphs to

drawing large and changing ones.

Variations. `sizeBy(spc, 1, 10)` (before `showOnly`) widens edges by their path count, showing the main path and its strongest tributaries.

Pitfalls. The main path is only as good as the paper set. It can't pass through work that the search missed. Citation cycles (papers citing each other's preprints) are ignored, so check `feedbackEdges()` if the path looks odd.

Name the research fronts Intermediate

You want the field's subareas, each with a name.

```
makeFromGDF("vis-papers.gdf")
removeDisconnected()
leidenClusters()
kw = readSets("vis-paper-keywords.csv", item="paper", set="keyword")
for grp in groupBy(leiden_class)[:6]:
    best = enrichment(grp, kw, show=False)[0]
    print(len(grp), best["set"], sortBy(grp, indegree)[-1].label)
```

How it works. Communities in a citation network are groups of papers that cite each other: research fronts. `enrichment` names each one by its most over-represented keyword (*Biology: Find modules and name them*), and the most cited member makes a good exemplar. The fronts found here are force-directed layout (around Fruchterman–Reingold), node-link diagrams and surveys (around Di Battista et al.'s book), planar and linear-time algorithms, aesthetics and symmetry (around Purchase's readability studies), hierarchical and Graphviz-style layout (around Gansner et al.), and dynamic network visualisation.

Co-authorship and keywords

Who writes with whom? Intermediate

You have a table of papers and their authors. **You want** the collaboration network.

Ingredients: `vis-paper-authors.csv` · `makeFromTwoMode` · `project`

```
makeFromTwoMode("vis-paper-authors.csv", "paper", "author")
crowded = [n for n in g.nodes if n.kind == "paper" and n.degree >= 20]
len(crowded)
for p in crowded: remove(p)
project(kind, "author")
removeDisconnected()
for n in sortBy(affiliations)[-5:]: print(n.label, n.affiliations, n.degree)
```

How it works. `makeFromTwoMode(file, "paper", "author")` reads one row per authorship as a two-mode network: paper and author nodes, with `kind` saying which. `project(kind, "author")` then links authors who wrote together (*SNA: Project onto one kind*). `affiliations` is each author's number of papers in the set. Peter Eades, Stephen Kobourov and Roberto Tamassia have the most.

Pitfalls. One paper with 25 authors adds 300 ties to the projection, and swamps the collaboration structure. Here, four such papers turn out to be atmospheric science: the title search for "force-

directed" matched "direct … forcing". Look at the outliers before projecting. They're where data problems hide. Removing papers with 20 or more authors fixes both problems.

Map the keywords Intermediate

```
makeFromTwoMode("vis-paper-keywords.csv", "paper", "keyword")
project(kind, "keyword", method="jaccard")
remove(affiliations < 8)
remove(weight < 0.05)
showOnly(giantComponent())
louvainClusters()
colorize(modularity_class)
sizeBy(affiliations, 12, 60)
labelBy(label)
labelSizeBy(affiliations)
forceAtlas2Layout()
```

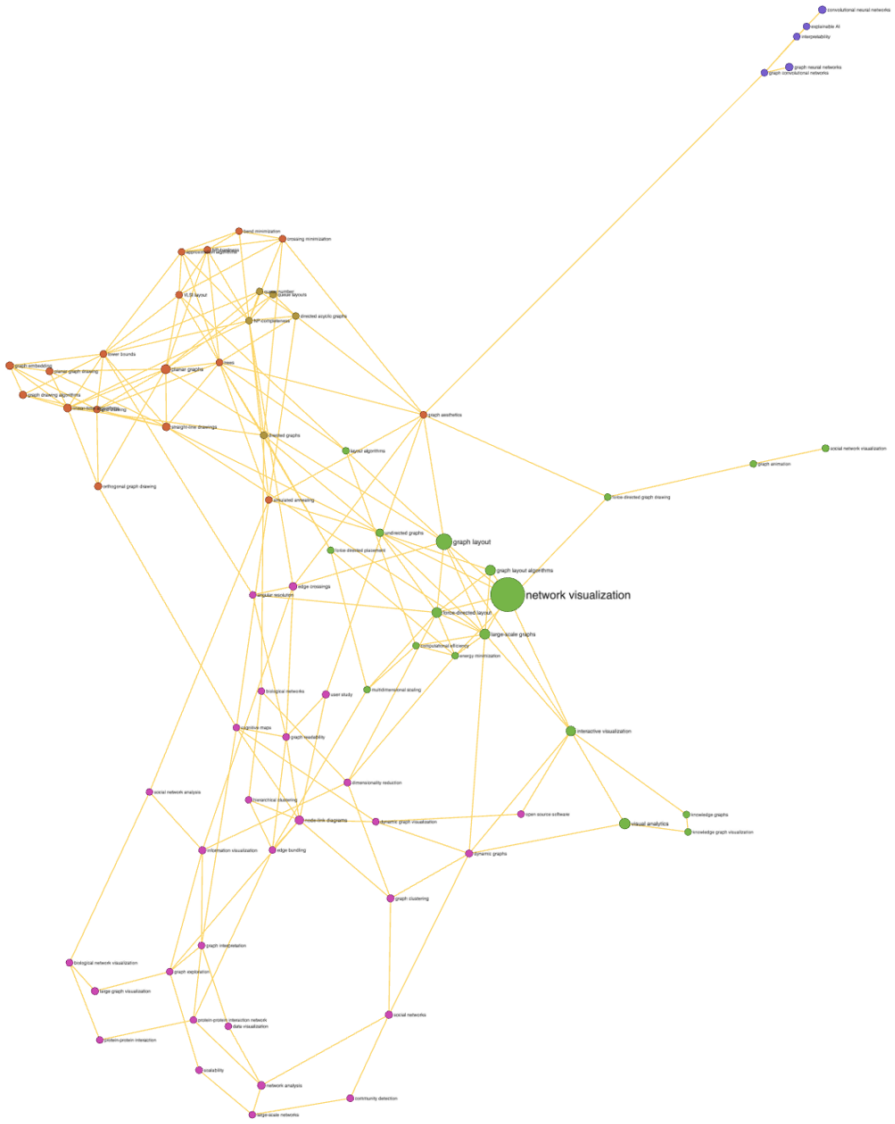


Figure 3: Keyword co-occurrence in graph-visualisation papers

How it works. Projecting papers × keywords onto the keywords links two keywords when papers share them. Jaccard weights (shared papers ÷ papers with either) keep a ubiquitous keyword like "network visualization" (382 papers) from linking to everything. Removing rare keywords (in fewer than 8 papers) and weak links leaves 86 keywords. The 73 that stay connected form a thematic map whose communities are the field's topics. Filter first, then keep the largest component, since each removal can strand more nodes.

Similarity

Link documents with similar content Intermediate

You have documents described by features (keywords, term counts, measurements). **You want** each document linked to the ones most like it, whether or not they cite each other.

Ingredients: `vis-paper-keywords.csv`, `vis-papers-nodes.csv` · `makeFromSimilarity`

```
makeFromSimilarity("vis-paper-keywords.csv", features="keyword", values="score",
 ↪  k=4)
makeFromCSV("vis-papers-nodes.csv")
leidenClusters()
kw = readSets("vis-paper-keywords.csv", item="paper", set="keyword")
for grp in groupBy(leiden_class)[:5]:
    print(len(grp), enrichment(grp, kw, show=False)[0]["set"])
```

How it works. `makeFromSimilarity` turns a table into a network: each row (here, each paper) is linked to its `k` most similar rows. The table can be *wide* (one column per feature) or, as here, *long*, with one row per paper, keyword and score (`features=` and `values=` name the columns). `method` chooses the similarity:

method	similar when	use for
`"cosine"` (default)	profiles point the same way	term weights, keyword scores
`"correlation"`	values rise and fall together	expression profiles, time series
`"jaccard"`	the same features are present	yes/no data
`"euclidean"`	values are close	measurements on one scale

`threshold=0.5` links every pair at least that similar instead of the top `k`, and `mutual=true` keeps only pairs that are in each other's top `k`. `makeFromCSV` then adds titles and years from the node table.

Pitfalls. A k-nearest-neighbour network links every node to *k* others, even an outlier with no real neighbours, so look at the weights (`plotHistogram(weight)`). A threshold network leaves outliers isolated, which is often more honest.

Knowledge graphs

Query Wikidata Intermediate

You want a network from a public knowledge graph, here "influenced by" links among philosophers.

```
q = """SELECT ?source ?sourceLabel ?target ?targetLabel WHERE {
  ?source wdt:P106 wd:Q4964182; wdt:P737 ?target .
  ?target wdt:P106 wd:Q4964182 .
  SERVICE wikibase:label { bd:serviceParam wikibase:language "en". }
} LIMIT 400"""
makeFromSPARQL(q)
sizeBy(indegree, 6, 40)
```

```
labelBy(label)
g.nodes.labelpos = "outside"
forceAtlas2Layout()
```

How it works. `makeFromSPARQL(query)` runs a SPARQL query (on Wikidata unless you give another endpoint) whose rows bind `?source` and `?target`. `?sourceLabel`/`?targetLabel` become labels, and any other variables become edge fields. In Wikidata, `P106` is occupation, `Q4964182` is philosopher and `P737` is "influenced by", so edges run from each philosopher to their influences. The most cited influences are Aristotle, Marx, Plato, Hegel and Descartes.

Pitfalls. Knowledge graphs are incomplete in uneven ways: famous, Western and recent subjects are better covered. `LIMIT` also truncates silently. Read the result as "what Wikidata records", and check the total count with a `COUNT(*)` query.

Case studies

Case study: the contest in "A Scandal in Bohemia" Basic

```
cast = {"Holmes": ["Holmes", "Sherlock"],
        "Irene Adler": ["Irene Adler", "Irene", "Miss Adler", "the woman", "Mrs.
        ↪ Norton"],
        "the King": ["King", "Majesty", "Count Von Kramm", "Von Kramm"],
        "Godfrey Norton": ["Godfrey Norton", "Godfrey", "Mr. Norton"]}
makeFromText("scandal-in-bohemia.txt", cast)
[(n.label, n.first, n.units) for n in sortBy(first)]
```

The `first` field gives the order of entrances. Irene Adler is named in the third paragraph (Watson's frame: she is "the woman"), long before the King appears (paragraph 47), and Godfrey Norton comes last. Holmes is tied to both the King and Irene Adler, and Norton only to her. The King and Irene Adler, whose quarrel is the story, are never named in the same paragraph, though he spends pages telling her story. In dialogue, speakers are rarely named, so the paragraphs where he speaks about her don't mention him. A co-mention network shows who is *named* together. Here that means the plot runs through Holmes, but conversations don't show up. For a play, where every speech is labelled with its speaker, link speakers by scene instead.

Case study: how graph drawing grew up Advanced

```
makeFromGDF("vis-papers.gdf")
removeDisconnected()
leidenClusters()
mp = mainPath()
addNodeField("decade", INTEGER, 0)
kw = readSets("vis-paper-keywords.csv", item="paper", set="keyword")
names = {}
for grp in groupBy(leiden_class)[:8]:
    names[grp[0].leiden_class] = enrichment(grp, kw, show=False)[0]["set"]
for n in g.nodes: n.decade = n.year // 10 * 10
```

```
for d in (1980, 1990, 2000, 2010):
    fronts = sorted(groupBy(decade == d, leiden_class), key=len)[-3:]
    top = [(names.get(f[0].leiden_class, "other"), len(f)) for f in
↪   reversed(fronts)]
    print(d, len(decade == d), top)
```

This puts the pieces together. Each research front is named by its top keyword, and the papers of each decade are counted by front. In the 1980s and 1990s the largest front is algorithmic ("linear-time algorithms": planar and orthogonal drawing), with layout second. By the 2000s, the force-directed "network visualization" front and the "node-link diagrams" front have caught up. In the 2010s node-link work (bundling, surveys, evaluation) leads. Overlay the main path (`mp.color = red`), which moves through the same fronts in the same order. One citation network and three recipes (communities, their names, the main path) give a defensible short history of a field. Check it against what its practitioners say, of course.